

CANDIDATE
NAME

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CENTRE
NUMBER

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CANDIDATE
NUMBER

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9618/32

October/November 2023

1 hour 30 minutes

You must answer on the question paper.

No additional materials are needed.

- Answer **all** questions.
- Use a black or dark blue pen.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use an HB pencil for any diagrams, graphs or rough working.
- Calculators must **not** be used in this paper.

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [].
- No marks will be awarded for using brand names of software packages or hardware.

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[Turn over

1 (a) Real numbers are stored in a computer using floating point representation with:

- 10 bits for the mantissa
- 6 bits for the exponent
- two's complement form for both the mantissa and the exponent.

Write the normalised floating-point representation of -96.75 in this system.

Show your working.

Mantissa

--	--	--	--	--	--	--	--	--	--

Exponent

--	--	--	--	--	--

Working

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[3]

(b) Explain why a binary representation is sometimes only an approximation to the real number it represents.

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[3]

- 2 Describe what is meant by **composite** and **non-composite** data types.

Composite

.....

.....

.....

Non-composite

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.....

.....

[4]

- 3 The location of a record in a random file is determined using a hashing algorithm.

A collision may occur during the process of adding a record.

- (a) Outline what is meant by the term **collision** in this context.

.....

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.....

[2]

- (b) Explain how a collision can be dealt with when writing records to a random file.

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[3]

- 4 Complete the following paragraph about a **protocol suite**, using words from the given list.

Some words are **not** used.

BitTorrent	circuit switching	layered	link	list
peer-to-peer	queue	stack	star	TCP/IP

The protocols in a determine the interconnectivity rules for a network model such as the model.

[3]

- 5 (a) Outline the reasons why an operating system may need to use virtual memory.

.....

 [2]

- (b) Explain the circumstances in which disk thrashing could occur.

.....

 [3]

- 6 (a) The Reverse Polish Notation (RPN) expression:

$$a \ b \ * \ 2 \ / \ c \ d \ / \ *$$

is to be evaluated where $a = 20$, $b = 3$, $c = 10$ and $d = 5$.

Show the changing contents of the following stack as the RPN expression is evaluated.

[4]

- (b) Explain how an expression stored in RPN can be evaluated.

.....

.....

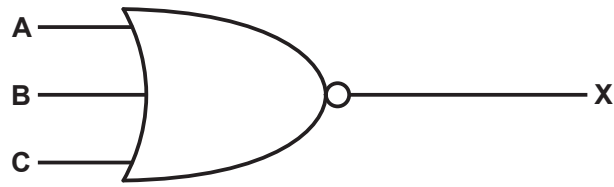
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..... [3]

- 7 (a) This logic circuit represents the Boolean expression: $X = \overline{A + B + C}$



Complete this truth table for the given logic circuit.

A	B	C	X
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

[1]

- (b) Apply De Morgan's laws to the expression: $X = \overline{A + B + C}$

$X =$ [1]

- (c) Simplify the following expression using Boolean algebra.

Show all the stages in your simplification.

$$T = X.Y.Z + X.\bar{Y}.Z + \bar{X}$$

.....

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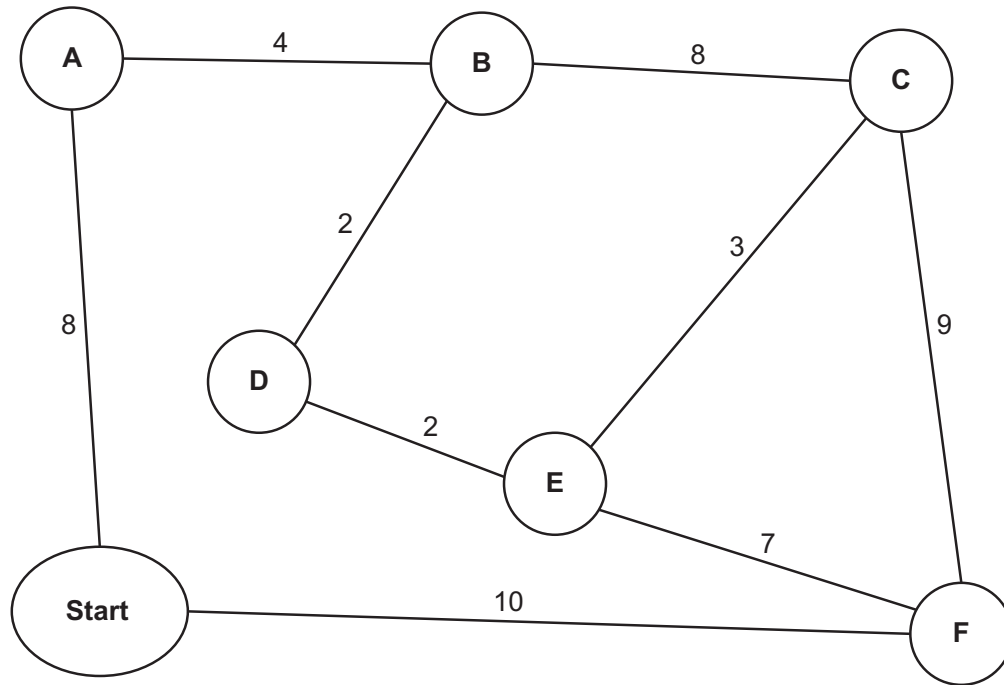
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..... [3]

- 8 Calculate the shortest distance between the **Start** and each of the destinations in the diagram using Dijkstra's algorithm.

Show your working **and** write your answers in the table provided.



Working

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Answers:

A	B	C	D	E	F

[5]

- 9 (a) A stack Abstract Data Type (ADT) is to be implemented using pseudocode, with procedures to initialise it and to push new items onto the stack.

A 1D array `Stack` stores the contents of the stack.

- (i) Study the pseudocode in **part (a)(ii)** and complete the table of identifiers by writing the missing data types and descriptions.

Identifier	Data type	Description
BasePointer		
TopPointer		
Stack	REAL	

[2]

- (ii) Complete the pseudocode.

```

CONSTANT MaxSize = 40
DECLARE BasePointer : INTEGER
DECLARE TopPointer : INTEGER
DECLARE Stack : ARRAY[1:40] OF REAL

// initialisation of stack
PROCEDURE Initialise()

    ..... ← 1

    ..... ← 0
ENDPROCEDURE

// push an item onto the stack
PROCEDURE Push(NewItem : REAL)

    ..... MaxSize THEN

    .....

    Stack[TopPointer] ← .....
ENDIF
ENDPROCEDURE

```

[5]

(b) Justify the use of a linked list instead of an array to implement a stack.

.....

.....

.....

..... [2]

(c) Explain how a compiler makes use of a stack when translating recursive programming code.

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..... [4]

10 Describe the features of the SIMD and MISD computer architectures.

SIMD

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.....

.....

MISD

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.....

..... [4]

- 11 A **declarative** programming language is used to represent some facts about people and their hobbies.

```

01 hobby(music) .
02 hobby(caving) .
03 hobby(climbing) .
04 hobby(camping) .
05 hobby(baking) .
06 hobby(travelling) .
07 person(toby) .
08 person(natasha) .
09 person(fatima) .
10 person(joseph) .
11 person(elijah) .
12 person(nina) .
13 enjoys(natasha, travelling) .
14 enjoys(toby, climbing) .
15 enjoys(nina, climbing) .
16 enjoys(elijah, camping) .
17 enjoys(fatima, baking) .
18 enjoys(joseph, camping) .
19 dislikes(toby, caving) .

```

These clauses have the meanings:

Clause	Meaning
01	Music is a hobby
07	Toby is a person
13	Natasha enjoys travelling
19	Toby dislikes caving

- (a) Carlos is a person who enjoys the hobby of cycling but does not like music.

Write additional clauses to represent this information.

```

20 .....
21 .....
22 .....
23 .....

```

[4]

(b) Using the variable *P*, the goal:

`enjoys(P, camping)`

returns

`P = elijah, joseph`

Write the result returned by the goal:

`enjoys(P, climbing)`

`P = .....` [1]

(c) *N* is a person who might enjoy *H* if *H* is a hobby and *N* does not dislike *H*.

Write this as a rule.

`might_enjoy(N, H)`

IF

 [4]

12 (a) Describe, with an example, what is meant by an **exception**.

.....

 [2]

- (b) A pseudocode algorithm searches for a customer record in a random file AccountRecord.dat. A user inputs the name of the customer.

The records are stored using the user-defined data type TAccount.

```
TYPE TAccount
    DECLARE AccountNumber : INTEGER
    DECLARE Name : STRING
    DECLARE Address : STRING
    DECLARE Telephone : STRING
ENDTYPE
```

If the record is found, it is output, otherwise an error message is displayed.

Complete the file handling pseudocode.

```
DECLARE Customer : TAccount
DECLARE Location : INTEGER
DECLARE MaxSize : INTEGER
DECLARE FoundFlag : BOOLEAN
DECLARE SearchCustomer : STRING
MaxSize ← 1000

OPENFILE .....
Location ← 1

..... ← FALSE
OUTPUT "Enter the customer's name"

.....

..... AND Location <= MaxSize

..... "AccountRecord.dat", .....
GETRECORD "AccountRecord.dat", Customer
IF SearchCustomer = Customer.Name THEN
    OUTPUT "Customer found: "
    OUTPUT Customer           // output customer record
    FoundFlag ← TRUE
ENDIF
Location ← Location + 1
ENDWHILE
IF NOT FoundFlag THEN

    OUTPUT "....."

ENDIF
```

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